Problem Session II

Question 1

A school club has 8 members. The club is sending 2 students to represent them at a regional event. One student will be assigned the role of team captain, and the other will be assigned the role of note-taker. Each role is different and must be filled by a different student.

How many different ways can the club assign these two roles?

A teacher has a class of 8 students and wants to form a study pair. Each pair consists of 2 students who will work together. How many different study pairs can be formed?

Question 2

Suppose $A \subseteq B$.

Let's prove that the P(A) $\leq$ P(B).

Question 3 (Homework-2 Question 2)

Show that for any two events A and B, we have $P(A \cap B) \geq P(A) + P(B) - 1$.

Question 4 (Homework-2 Question 6)

Before attempting to answer the following questions, perform the steps outlined below using the seed 1284:

– Generate a set containing the numbers 1 to 50000, and store that set in an object called 'Set1'.

– Using the sample() function, generate a set that contains values from Set1 in no particular order, allowing for repeats. Store this set in an object called Omega.

– Generate three subsets of Omega of size 100. Each individual subset is to be made without replacement, but each individual subset should sample from the full Omega. Label these sets A, B and C, in that order.

– To acheive the intended subsets, once you have written the appropirate code, run it all at the same time.

(a) For each of the following, fill in the blank with the most appropriate symbol from: , (5 points)

i. 50000 ___ Ω

ii. 50000 ___ C

iii. 48231 ___ Ω

iv. 48231 ___ A

v. 2707 ___ B

(b) Do the sets generated, intersect? If so, state specifically which sets and what elements fall in the intersection.